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CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– FPT University, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
27/10/2026	M	Lê Nguyễn Quang Đức	Expand use case and feature tables to all 8 mandatory FRs, remove NFR template placeholder text, and update ERD baseline reference to 23 pages (12 ERDs).

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

VISTA is a multi-role education platform for schools, teachers and learners, linking authorized classroom robot allocation to learning assignment, versioned student work, schematic experiment requests and teacher feedback. Local Robot Runtime and AI integration remain separate trust boundaries; physical E2E must be verified before marking implemented.

VISTA is a role-governed robotics education platform. Roles: System Admin, School Admin, Instructor/Teacher and Student. External integrations are Supabase Auth/PostgreSQL, optional AI/model service and local safety-managed Robot Runtime. System Context is in the supplied C4 diagrams; only code-verified responsibilities are mandatory.

Learning loop: Learn → Predict → Build → Experiment → Reflect → Assess. Classroom resource allocation grants access; it does not create a live robot control session.

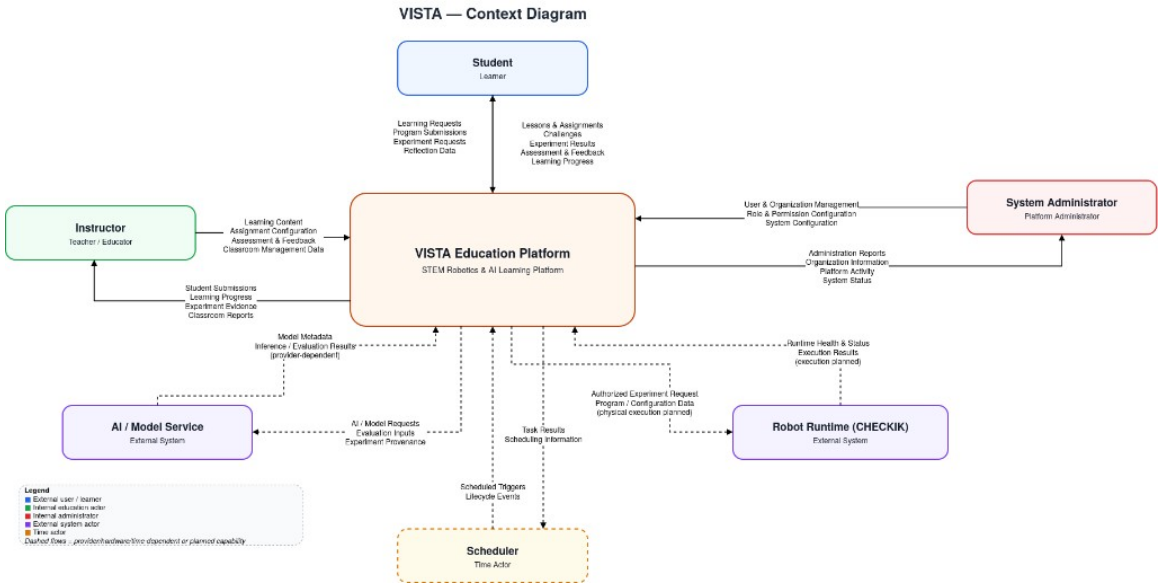


Figure 1. VISTA Platform — System Context Diagram

2. User Requirements

School Admin manages classroom and entitled robot allocation; Teacher publishes assignments and reviews student work; Student accesses own class, versions work and requests permitted experiments; system/external actors include AI/Model Service and Local Robot Runtime.

2.1 Actors

Authorization derives JWT subject → public users → active organization and classroom membership. Scope permission is required alongside active membership; out-of-scope resources may return 404.

- Students: their classes, authorized assignments, programs, experiments, feedback and reflections.
- Teachers: class delivery, assignments, review and assessments within teacher/TA classroom relationship.
- School administrators: school members, classrooms, entitled robots and classroom allocations.

System administrators: globally scoped role/organization administration only with authoritative permission.

#	Actor	Description
1	System Administrator	Platform-scoped administration; verify code roles
2	School Administrator	Own school classrooms and robot allocation
3	Teacher	Publish classroom learning and review authorized work

2.2 Use Cases

UC-01 school admin allocates an entitled robot; UC-02 teacher publishes class assignment and student submits immutable program; UC-03 student runs schematic preview and symbolic evaluation; UC-04 teacher assesses student work and student reflects.

2.2.1 Diagram(s)

Relevant source diagrams supplied in VISTA_DIAGRAM.pdf: pages 1–7 for architecture and 4 workflow activities. This PDF is referenced here; rendered design figures are embedded in Report 4.

Use Case Diagram is an SRS illustration; linked C4 and activity diagrams should be updated from true route and schema. No completed physical robot execution claim.

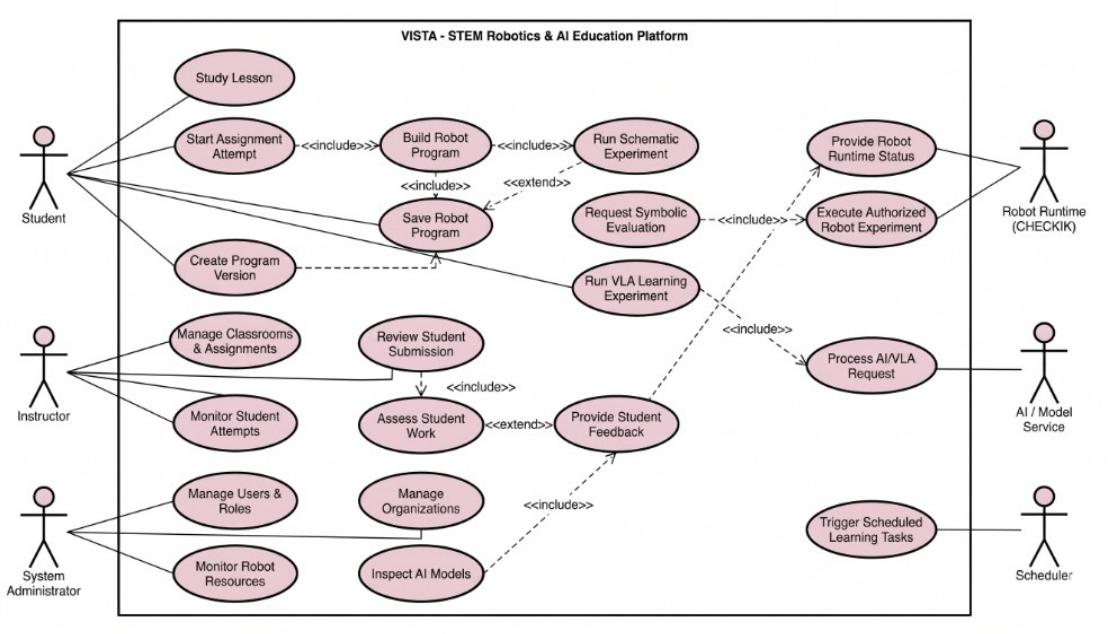


Figure 2. VISTA Platform — Use Case Diagram

2.2.2 Descriptions

UC main/exception scenarios and owners are in BM01 Appendix A. Acceptance requires persisted entity state and scoped authorization checks.

ID	Use Case	Actors	Use Case Description
UC-01	Authentication and role-scoped access	System Admin / School Admin / Teacher / Student	Requests require valid user, active membership, and scoped authorization; cross-organization access is denied.
UC-02	Classroom robot allocation and governance	School Admin	Only an entitled robot can be allocated to an owned classroom; conflicting active allocations are atomically rejected.
UC-03	Lesson preparation and assignment publication	Teacher	Authorized teacher creates and publishes classroom assignments; unauthorized publish requests are rejected.
UC-04	Immutable program versioning and submission	Student	Learner edits RobotProgram drafts, generates immutable versions, and submits an eligible version for an active attempt.
UC-0	Schematic experiment execution and symbolic	Student	Student requests a schematic experiment run; backend

5	evaluation		validates eligibility and persists truthful execution state and evidence.
UC-06	Bounded physical robot execution through local runtime	Student, Robot Runtime	Physical execution requires active reservation, hardware readiness, and safety checks; failure or stop fails closed.
UC-07	Teacher assessment, feedback, and student reflection	Teacher, Student	Teacher reviews submission evidence and records rubric assessment; student inspects feedback and records reflection.
UC-08	AI model and dataset provenance metadata inspection	Student / Teacher / System Admin	Users view traceable model versions, training datasets, and evaluation metrics; external writes fail gracefully.

3. Functional Requirements

3.1 System Functional Overview

The functional baseline encompasses eight mandatory functional requirements (FR-01 to FR-08) structured across four core workflows (WF-01 to WF-04). Scope remains consistent with Registration v1.2 and verified API authorization. Expected error flows include unauthorized organization or classroom scope, invalid resource entitlement, foreign submission tampering, and unsupported execution modes.

Functional requirements must be independently testable and mapped to implemented controllers/services. Examples below separate implemented API capabilities from planned physical execution.

3.1.1 Screens Flow

School Admin: organization/classroom -> eligible robot -> allocation; Teacher: lesson -> assignment -> publish -> review; Student: assigned content -> versioned program -> submit -> schematic experiment -> view feedback. UI routes require code-level confirmation.

Student → course/lesson → own assignment → own attempt → program version → experiment → submission and feedback. Teacher and school administration routes are independently permission-scoped.

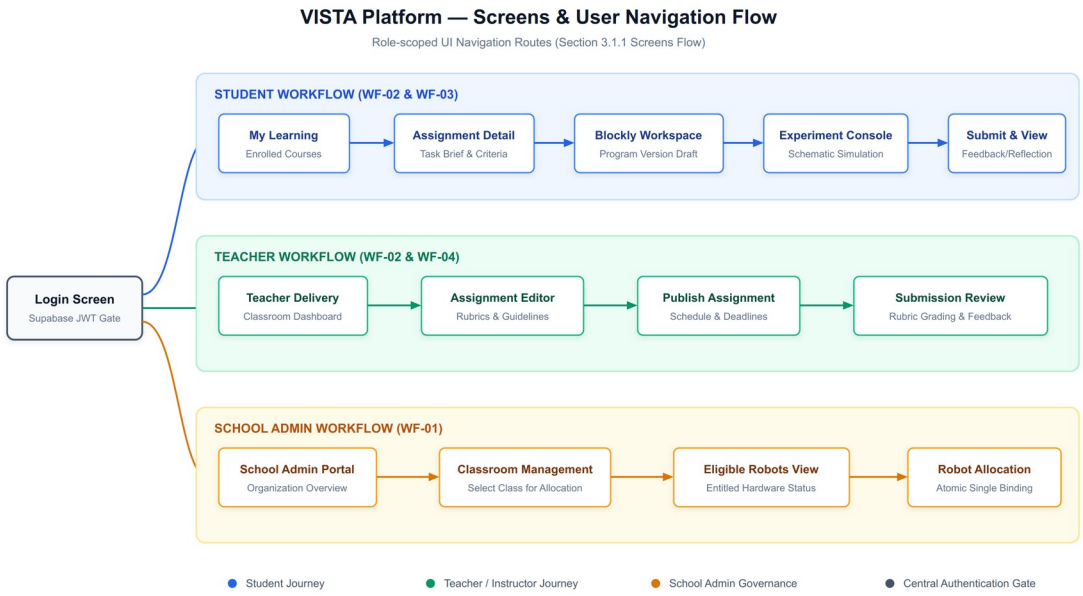


Figure 3. VISTA Platform — Screens & User Navigation Flow

3.1.2 Screen Descriptions

Access, School/Classroom Management, Learning Content and Assignment, Student Program/Submission, Experiment Request/Result, Teacher Assessment/Feedback. Detailed routes and controls are to be verified against web/robot-edu; conceptual flow is not evidence of a working page.

Screens: Login, School Admin Classrooms/Robot Allocation, Teacher Course/Assignment, Student Program/Experiment, Instructor Assessment, Student Feedback/Reflection.

#	Feature	Screen	Description
1	Role-governed access & identity gate	Login & Access Gate	JWT authentication, organization context selection, and scope validation.
2	Classroom robot allocation & governance	School Admin Dashboard	School-scoped classroom management and atomic robot allocation.
3	Assignment creation & delivery	Teacher Course / Delivery View	Classroom assignment authoring, scheduling, and publication.
4	Visual programming & version submission	Student Workspace / Blockly	Block-based coding, canonical AST compile, immutable versioning, and submission.
5	Schematic	Experiment	Schematic execution simulation, trajectory

	experiment & symbolic evaluation	Console	verification, and artifact persistence.
6	Bounded physical robot execution	Hardware Telemetry Monitor	Local device gateway status, readiness check, and supervised run dispatch.
7	Teacher assessment & student reflection	Assessment & Feedback Portal	Rubric grading, feedback delivery, and student reflection records.
8	AI model & dataset metadata inspection	Model / Dataset Registry	Traceable AI model cards, dataset versions, and benchmark evaluation metrics.

3.1.3 Screen Authorization

System Admin (platform scope), School Admin (organization), Teacher (assigned classroom/course), Student (own membership and work). Backend must check Supabase JWT, public.users mapping, active membership and per-resource scope.

Student cannot enter another school's classroom or assignment; teacher/TA requires active class relationship plus learning.delivery.manage; admin cannot authorize physical motion merely by allocating robot.

Screen	System Admin	School Admin	Teacher	Student
Login	Y	Y	Y	Y
School Resource Allocation	N	Y	N	N
Manage Assignments	N	N	Y	Read
Program Submission	N	N	Review	Y
Experiment Schematic	N	N	Review	Y
Assessment	N	N	Y	Read
Add New Data			X	X
Update All Data				X
Update Own Data				X
Update Managed Data				X
Delete Data				
…				

3.1.4 Non-Screen Functions

NestJS API handles authorization, resource entitlement checks, durable program versions/submissions, schematic experiment request persistence and teacher assessment; Python runtime device execution is an independent integration boundary.

Non-screen functions: JWT guard, scope service, Postgres advisory transaction locking for experiment deduplication, immutable ProgramVersion and safety-limited runtime adapter.

#	Feature	System Function	Description
1	Authentication & Access Control	FR-01	System verifies Supabase JWT and enforces strict organization and classroom scope guards on all endpoints.
2	Classroom Robot Allocation	FR-02	School Admin allocates an entitled robot to an owned classroom; atomic replacement prevents dual active allocations.
3	Assignment Publication	FR-03	Teacher publishes classroom work; unauthorized actors or foreign teachers cannot publish or alter assignments.
4	Versioned Program Submission	FR-04	Student creates immutable program versions and submits an attempt; other students cannot submit or alter work.
5	Schematic Experiment Execution	FR-05	Student runs schematic simulation; backend persists deterministic execution logs, status, and evaluation results.
6	Physical Execution Boundary	FR-06	System dispatches physical commands only to ready, reserved hardware; duplicate or unsafe commands fail closed.
7	Assessment & Feedback Delivery	FR-07	Teacher submits score and feedback for student submission; student receives feedback and submits reflection.
8	AI / Model Metadata Management	FR-08	Platform serves read-only metadata for AI model cards, dataset lineage, and evaluation metrics with zero false claims.

3.1.5 Entity Relationship Diagram

Logical ERD reference: pages 12–23 of VISTA_DIAGRAM.pdf; diagrams contain proposed cardinalities and require verification against Prisma/migrations including PK, FK, nullable and unique constraints.

Master Logical ERD is a curated view from Supabase columns/FKs; compare all FK constraint groups with nullability/unique constraints; no arbitrary extra relationships.



Figure 4. VISTA Platform — Logical ERD (School & Access Scope)

Entities Description

#	Entity	Description
1	learning_classroom_robot_allocations	Scoped robot authorization
2	learning_classroom_assignments	Published classroom task
3	learning_submissions	Student work version link
4	learning_experiments	Experiment request and status

3.2 WF-01 School Administration and Classroom Robot Allocation

FR-02: School Admin selects own classroom and a currently entitled robot. API verifies active role/organization/classroom membership and entitlement; it upserts allocation atomically. Postcondition is an authorization record, not a live robot control session. Reject cross-school/expired-entitlement requests.

3.2.1 School Admin Allocates Robot

Precondition: authorized school admin, existing classroom in organization, eligible robot. Main: list classroom and eligible robots; select and submit; guard validates scope; service records one classroom allocation; display current allocation. Exception: 403/404 unauthorized scope; conflict/validation failure returns without unauthorized change.

Trigger: School Admin selects an own classroom and an entitled robot.

- Validation: authenticated organization permission, active school membership, classroom ownership, entitlement and allocation conflict rules.
- Result: one classroom robot allocation; atomic replace. Allocation is not an execution lease.

- API: GET /api/v1/education-admin/classrooms/:classroomId/eligible-robots; PUT /api/v1/education-admin/classrooms/:classroomId/robot-allocation.
- Failure: inaccessible class, no entitlement and conflict are rejected without cross-school data leakage.

3.3 WF-02 Learning Content and Assignment Management

FR-03 & FR-04: Teacher manages and publishes classroom assignments; Student accesses learning lessons, develops visual robot programs with immutable versioning, and submits assignment attempts. All actions are scoped to verified classroom enrollment.

3.3.1 Teacher Publishes Assignment

Precondition: authenticated teacher with active classroom assignment role. Main: teacher defines assignment criteria, rubric guidelines, and submission deadlines; publishes assignment to enrolled students. Exception: unauthorized teacher access or foreign classroom modification rejected.

Trigger: Teacher creates and confirms classroom assignment publication.

• Validation: active teacher membership, ownership of target classroom, valid rubric criteria definition.
• Result: assignment status transitions to PUBLISHED; enrolled students receive access to task brief.
• API: POST /api/v1/education/classrooms/:classroomId/assignments; GET /api/v1/education/assignments/:id.
• Failure: unassigned teacher or invalid classroom returns 403 Forbidden.

3.3.2 Student Creates and Submits Program Version

Precondition: authenticated student enrolled in classroom with an active published assignment. Main: student opens visual Blockly workspace, writes robot program, saves immutable versions with SHA-256 integrity hash, and submits version for assignment attempt. Exception: submission after deadline without extension or cross-student attempt tampering rejected.

Trigger: Student clicks "Submit Version" for an active attempt.

• Validation: active student enrollment, attempt ownership, program AST syntactic validity.
• Result: immutable program version created; submission record logged with status SUBMITTED.
• API: POST /api/v1/education/programs/:programId/versions; POST /api/v1/education/submissions.
• Failure: foreign attempt modification or empty program AST returns 400/403.

3.4 WF-03 Robot Experiment Execution

FR-05 & FR-06: Student requests experiment run against submitted program version. Education API executes deterministic schematic simulation and records execution artifacts. Physical robot execution via CHECKIK runtime requires explicit reservation and verified hardware safety readiness.

3.4.1 Schematic Experiment Simulation

Precondition: authenticated student with valid program version. Main: student initiates schematic run; backend validates AST schema, executes deterministic kinematic simulation, persists trajectory coordinates and score evidence. Exception: invalid AST or unsupported execution mode returns validation error.

Trigger: Student requests schematic execution in Experiment Console.

- Validation: student submission ownership, supported executionMode (schematic), validated program AST.
- Result: experiment record created; deterministic simulation executed; trajectory artifact persisted with status COMPLETED.
- API: POST /api/v1/education/experiments; GET /api/v1/education/experiments/:id.
- Failure: unsupported execution mode returns 400 Bad Request.

3.4.2 Physical Robot Execution Boundary

Precondition: active robot reservation, verified safety boundary, and connected CHECKIK robot runtime. Main: authorized dispatch service sends bounded pick-and-place command sequence to local robot runtime; receives telemetry and execution evidence. Exception: safety trigger or hardware disconnect aborts execution immediately (fail-closed).

Trigger: Authorized run request dispatched to reserved physical robot.

- Validation: active reservation window, hardware calibration check passed, safety profile active.
- Result: physical robot performs bounded tabletop manipulation; runtime returns execution telemetry.
- API: POST /api/v1/robot-runtime/dispatch; GET /api/v1/robot-runtime/telemetry.
- Failure: hardware fault or emergency stop triggers immediate halt and records safety event.

3.5 WF-04 Assessment, Feedback and AI Integration

FR-07 & FR-08: Authorized teacher assesses student submission, scores rubric criteria, and delivers qualitative feedback. Student reviews feedback and logs learning reflections. Platform serves traceable read-only AI model metadata and training dataset lineage.

3.5.1 Teacher Assessment and Rubric Scoring

Precondition: authenticated teacher assigned to classroom with student submission in SUBMITTED state. Main: teacher reviews submission evidence, executes comparison runs, assigns scores per rubric criterion, and publishes written feedback. Exception: foreign teacher access or grading without assignment scope rejected.

Trigger: Teacher submits completed assessment for student submission.

- Validation: teacher classroom authorization, submission ownership, valid score range per criterion.
- Result: assessment record persisted; submission transitions to ASSESSED; student reflection unlocked.
- API: POST /api/v1/education/submissions/:submissionId/assessment; GET /api/v1/education/assessments/:id.
- Failure: unauthorized teacher returns 403 Forbidden.

3.5.2 Student Reflection and AI Metadata Inspection

Precondition: authenticated student with assessed submission; or authorized user inspecting AI models. Main: student enters qualitative reflection notes based on teacher feedback; user inspects AI model cards, training datasets, and performance evaluation metrics. Exception: reflection on unassessed submission rejected.

Trigger: Student submits reflection notes or user views AI model registry.

- Validation: student attempt ownership; valid model card identifier.
- Result: student reflection saved to submission history; AI model metadata and lineage displayed.
- API: POST /api/v1/education/submissions/:submissionId/reflection; GET /api/v1/ai-models.
- Failure: invalid submission ID or unauthenticated request returns 401/404.

4. Non-Functional Requirements

NFR acceptance targets must be measurable. Cross-school authorization isolation: zero unauthorized successful responses in a seeded negative-case suite. Defense 1.1 targets (not current results): mandatory main scenarios 100%; exception scenarios at least 80%, total cases pass at least 90%, >=50% of test cases automated CI.

4.1 External Interfaces

Web app HTTPS JSON -> Education API, Supabase Auth JWT, Prisma/PostgreSQL and evidence storage. AI service and Python Robot Runtime integration are design interfaces; confirm routes and deployment before asserting live E2E.

External interfaces: Supabase Auth JWT, NestJS /api/v1 REST/JSON, Prisma/PostgreSQL, external AI/model adapter (verify configuration), private Python Robot Runtime.

4.2 Quality Attributes

NFRs are proposed and require owner-confirmed test threshold, measurement environment and reporting; do not invent performance measurements.

4.2.1 Usability

Usability: a teacher and student must complete approved main workflow using clear actionable error messages; measure by timed usability session in staging.

4.2.2 Reliability

Reliability: authorization and state checks fail closed on unavailable dependencies; verify with fault injection and negative tests.

Availability: The VISTA Learning Platform targets ≥99.0% service availability during scheduled school instructional hours (07:00–18:00 UTC+7).

Mean Time Between Failures (MTBF): Targeted MTBF is ≥300 operating hours between unscheduled API or database service interruptions.

Mean Time To Repair (MTTR): Targeted MTTR is ≤2 hours for containerized API services in local and staging environments.

Accuracy: Robot program AST compilation and schematic trajectory evaluations are deterministic (100% precision with identical inputs).

Maximum Bugs or Defect Rate: Zero critical defects (data loss, privilege escalation, or unauthorized physical motion) permitted in release baselines.

Defect Severity Thresholds: Critical defects must be resolved within 24 hours; major defects within 5 business days; minor defects scheduled by sprint.

4.2.3 Performance

Performance: latency target and simultaneous user load to be signed off in BM01, measured with staged repeatable load test and recorded percentiles.

Response Time: Ordinary authenticated REST API operations return within P95 ≤3.0 seconds under nominal classroom concurrency (30 concurrent users).

Throughput: Education API handles a minimum of 50 requests per second across typical reading, submitting, and evaluation workloads.

Capacity: The database and API scale to support at least 1,000 active students and 10,000 immutable program versions per registered school.

Resource Utilization: API container memory utilization remains below 70% under standard instructional loads (512 MB memory limit).

4.2.4 Security & Safety

Additional: privacy/scope isolation and robot safety checks take priority over throughput.

5. Requirement Appendix

Appendix maps each FR to WF, owner and test IDs in the project test workbook; unverified state labels require correction against code.

5.1 Business Rules

BR-01 student sees only authorized classroom content and own attempts. BR-02 allocation conveys classroom authorization not active physical robot control. BR-03 physical control needs separate exclusive, safe runtime session. BR-04 experiment creation in observed Education API only supports schematic and creates pending record. BR-05 assessment changes require appropriate reviewer membership.

BR-01 Student reads only currently authorized classroom content and own attempts.

ID	Rule Definition
BR-01	Cross-organization reads/writes forbidden
BR-02	Classroom allocation is not runtime control
BR-03	Schematic mode only on observed Education API
BR-04	Program versions referenced by submission immutable
BR-05	Physical execution requires separate safety gate
BR-12	Order price is calculated as the sum of each food item price times the quantity of that food item ordered, plus applicable sales tax, plus a delivery charge if a meal is delivered outside the free delivery zone.
BR-24	To verify
BR-33	Network transmissions that involve financial information or personally identifiable information require 256-bit encryption.
BR-86	Only regular employees can register for payroll deduction for any company purchase.
BR-88	An employee can register for payroll deduction payment of cafeteria meals if no more than 40 percent of his gross pay is currently being deducted for other reasons.

5.2 Common Requirements

BR-02 Classroom robot allocation does not authorize physical motion; separate robot.control and safety checks are necessary.

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	AUTH_SCOPE	Error	Role/scope or input	Request outside authorized organization/classroom
2	ENTITL	Error	Role/scope or input	Robot not entitled to

	EMENT			this organization
3	EXPERIMENT_MODE	Error	Role/scope or input	Unsupported executionMode; only schematic allowed
4	SUBMISSION_OWNER	Error	Role/scope or input	User cannot modify another learner submission
5	MSG05	Toast message	Confirming email of asset hand-over is sent successfully	*A confirmation email has been sent to {email_address}.*
6	MSG06	Toast message	Resetting asset information successfully	*Return asset(s) successfully.*
7	MSG07	Toast message	Deleting asset information successfully	*Delete asset(s) successfully.*
8	MSG08	In red, under the text box	Input value length > max length	*Exceed max length of {max_length}.*
9	MSG09	In line	Username or password is not correct when clicking sign-in	*Incorrrect user name or password. Please check again.*

All user-visible success states must correspond to authoritative persisted data and runtime evidence.

Requirement-to-design cross-reference — revised ERD

ERD baseline: diagramall.drawio.xml, 23 pages containing 12 ERDs. Conceptual ERD (page 12) outlines core business entities; five Logical ERDs (pages 13–16b, including separate 16a Programs & Assessment and 16b AI & Datasets) define normalized attributes and foreign keys; six Physical ERDs (pages 17–22) establish PostgreSQL data types, primary/foreign key constraints, and unique indexes.